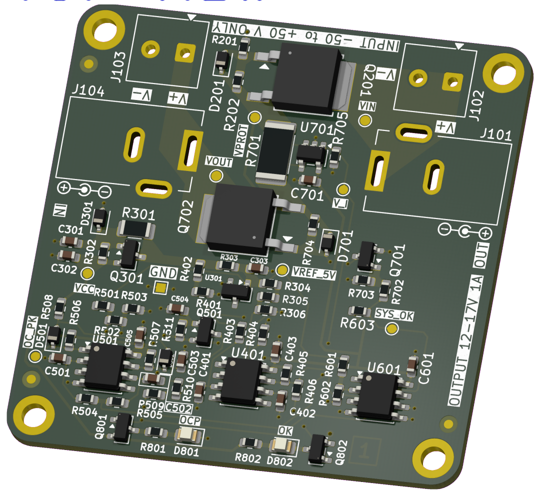
INPUT PROTECTION

2026-03-06

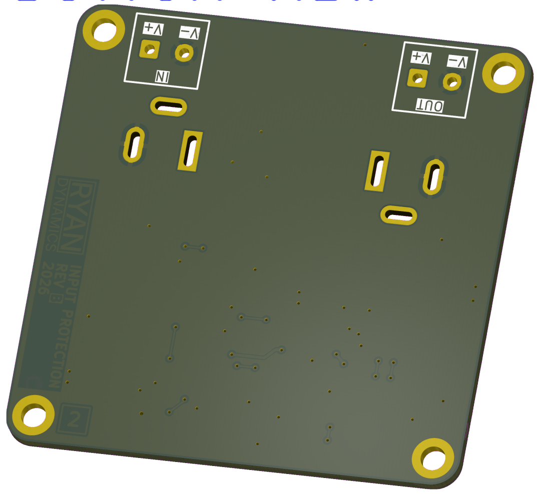
VARIANT: RELEASED

REV 1.0.0

TOP VIEW



BOTTOM VIEW



NOTES

COMMENT

NOT FITTED COMPONENTS MARKED AS X

DRAFT – EARLY STAGE SCHEMATIC CAPTURE
PRELIMINARY – LATE STAGE SCHEMATIC CAPTURE
CHECKED – PCB LAYOUT CHECKED, CONTACT ENGINEER IF MISTAKE FOUND
RELEASED – READY FOR PRODUCTION

DESIGN CONSIDERATIONS

DESIGN NOTE: Example text for informational design notes.

DESIGN NOTE: Example text for debug notes.

DESIGN NOTE: Example text for cautionary design notes.

DESIGN NOTE: Example text for critical design notes.

LAYOUT NOTE: Example text for critical layout guidelines.

PROJECT ASSIGNMENT 2	Ryan Dynamics		
DRN 2025-01-12 R. HICKS	THIS DESIGN AND/OR DRAWING IS THE PROPERTY OF RYAN DYNAMICS AND SHALL NOTBE REPRODUCED WITHOUT AUTHORISATION.		
CHK			
ENG APP	INPUT PROTECTION		
MFR APP			
SHEET PATH /	GIT HASH eebc9f1	DRAWING No	A3
FILENAME Input-Protection.kicad_sch	VARIANT RELEASED	REVISION 1.0.0	SHEET 1 OF 3

[2] SCHEMATIC

[1] https://www.ti.com/lit/an/slvae57b/slvae57b.pdf?ts=1770319054914

DESIGN NOTE:
This PMOS is acting as an ideal diode [1].

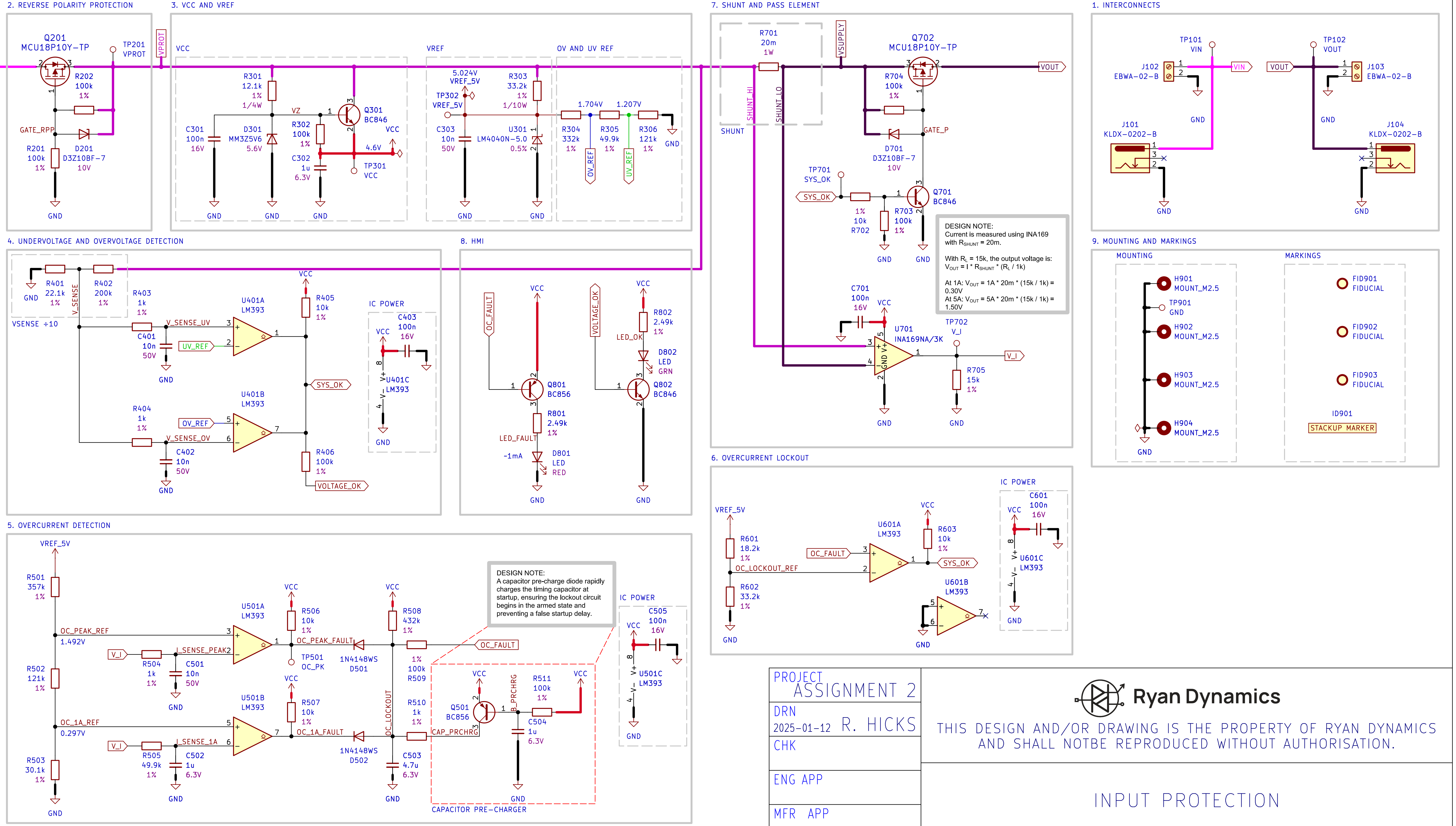
A 10 V zener from gate to source limits $V_{GS} \leq 10$ V under all VIN and transient conditions to protect the PMOS. A 100k gate to source resistor stops the gate floating. A 100k gate pulldown resistor sets default state to OFF when no voltage at VIN is present.

LAYOUT NOTE:
Route HI LO leads short from shunt to current amplifier.

DESIGN NOTE:
The over-current trip points are 1A and 5A, with 20m shunt.

At 1A, $V_{SHUNT} = I * R_{SHUNT} = 1A * 20m = 20mV$
At 5A, $V_{SHUNT} = I * R_{SHUNT} = 5A * 20m = 100mV$

LAYOUT NOTE:
We'll size for 5A current on 35um trace. We'll pick 3.5mm for a 25% margin. Keep traces short and wide, use pours/puddles when possible.



DESIGN NOTE:
Current is measured using INA169 with R_{SHUNT} = 20m.

With R_L = 15k, the output voltage is:
$V_{OUT} = I * R_{SHUNT} * (R_L / 1k)$

At 1A: $V_{OUT} = 1A * 20m * (15k / 1k) = 0.30V$
At 5A: $V_{OUT} = 5A * 20m * (15k / 1k) = 1.50V$

DESIGN NOTE:
A capacitor pre-charge diode rapidly charges the timing capacitor at startup, ensuring the lockout circuit begins in the armed state and preventing a false startup delay.

DESIGN NOTE:
The lockout delay is set by an RC network with R = 432k and C = 4.7u.

The time constant is: $\tau = R * C \approx 432k * 4.7u \approx 2s$

After a fault the capacitor must recharge before the circuit rearms, creating a ~2s lockout period.

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ENG APP	INPUT PROTECTION		
MFR APP			
SHEET PATH /SCHEMATIC/	GIT HASH eebc9f1	DRAWING No SCHEMATIC	A3
FILENAME Schematic.kicad_sch	VARIANT RELEASED	REVISION 1.0.0	SHEET 2 OF 3

[3] REVISION HISTORY

Version 1.0.0 – 2026-03-06

Fixed

- N/A

Added

- Schematic and PCB files

Changed

- N/A

Removed

- N/A

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CHK			
ENG APP	INPUT PROTECTION		
MFR APP			
SHEET PATH /REVISION HISTORY/	GIT HASH eebc9f1	DRAWING No REVISION HISTORY	A4
FILENAME Revision History.kicad_sch	VARIANT RELEASED	REVISION 1.0.0	SHEET 3 OF 3